

Comparison technique for semilinear elliptic problems on Riemannian Manifolds

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The Set Up

- Let (M^n, g) be a Riemannian manifold such that there exists $k \in \mathbb{R}$,
 $\text{Ric}_g \geq k(n-1)g$.
- Let $\Omega \subset M^n$ be a paracompact domain with $\mathcal{C}^2$ boundary.
- Let $f \in \text{Lip}_{\text{loc}}(\mathbb{R})$ be a locally Lipschitz function such that

$$f(x) > 0 \quad \text{and} \quad f(x) \geq nkx + f(0) \quad \text{for } x \in \mathcal{I}_f,$$

with $f(0) > 0$ when $k \leq 0$. Then, there exist non-empty connected intervals

$$0 \in \mathcal{R}_f \subset [0, \bar{r}_k) \quad \text{and} \quad \mathcal{I}_f \subset \mathbb{R}_+ \text{ with } 0 \in \partial \mathcal{I}_f.$$

Then we consider the following **semilinear elliptic problem**,

$$\begin{cases} \Delta u + f(u) = 0 & \text{in } \Omega, \\ u > 0 & \text{in } \Omega, \\ u = 0 & \text{in } \partial \Omega. \end{cases}$$



Basis for the Comparison

Model manifold

$$(M_k^n(F), g_k) := (I_k \times F, dr^2 + s_k(r)^2 g_F).$$

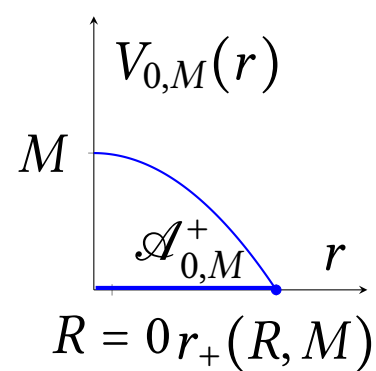
The Laplace–Beltrami operator on a model manifold,

$$\Delta = \partial_r^2 + (n-1) \cot_k(r) \partial_r + \frac{1}{s_k(r)^2} \Delta_F,$$

Model Solutions (*Radial solutions*)

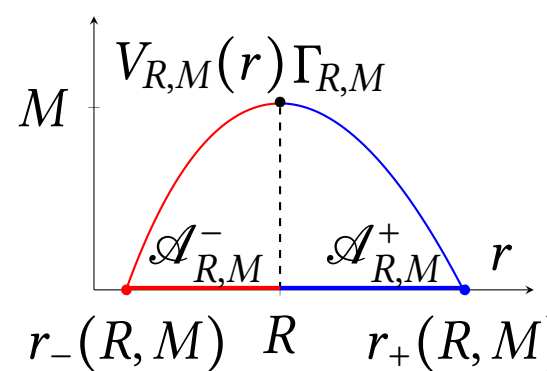
$$v(p) = V(r(p)) \quad \forall p \in \mathcal{A} \Leftrightarrow \begin{cases} V''(r) + (n-1) \cot_k(r) V'(r) + f(V(r)) = 0, \\ V(R) = M, \quad V'(R) = 0. \end{cases} \quad (\odot)$$

For each initial data (R, M) , we have two types of pairs solutions-domains, $(v_{R,M}, \mathcal{A}_{R,M})$, called **model pairs** and the family of solutions is denoted by Model_M ,



$$R = 0r_+(R, M)$$

$$\mathcal{A}_{0,M} = \{p \in M_k^n(F) : r(p) < r_+(0, M)\}$$



$$r_-(R, M) \quad R \quad r_+(R, M)$$

$$\mathcal{A}_{R,M} = \{p \in M_k^n(F) : r_-(R, M) < r(p) < r_+(R, M)\}$$

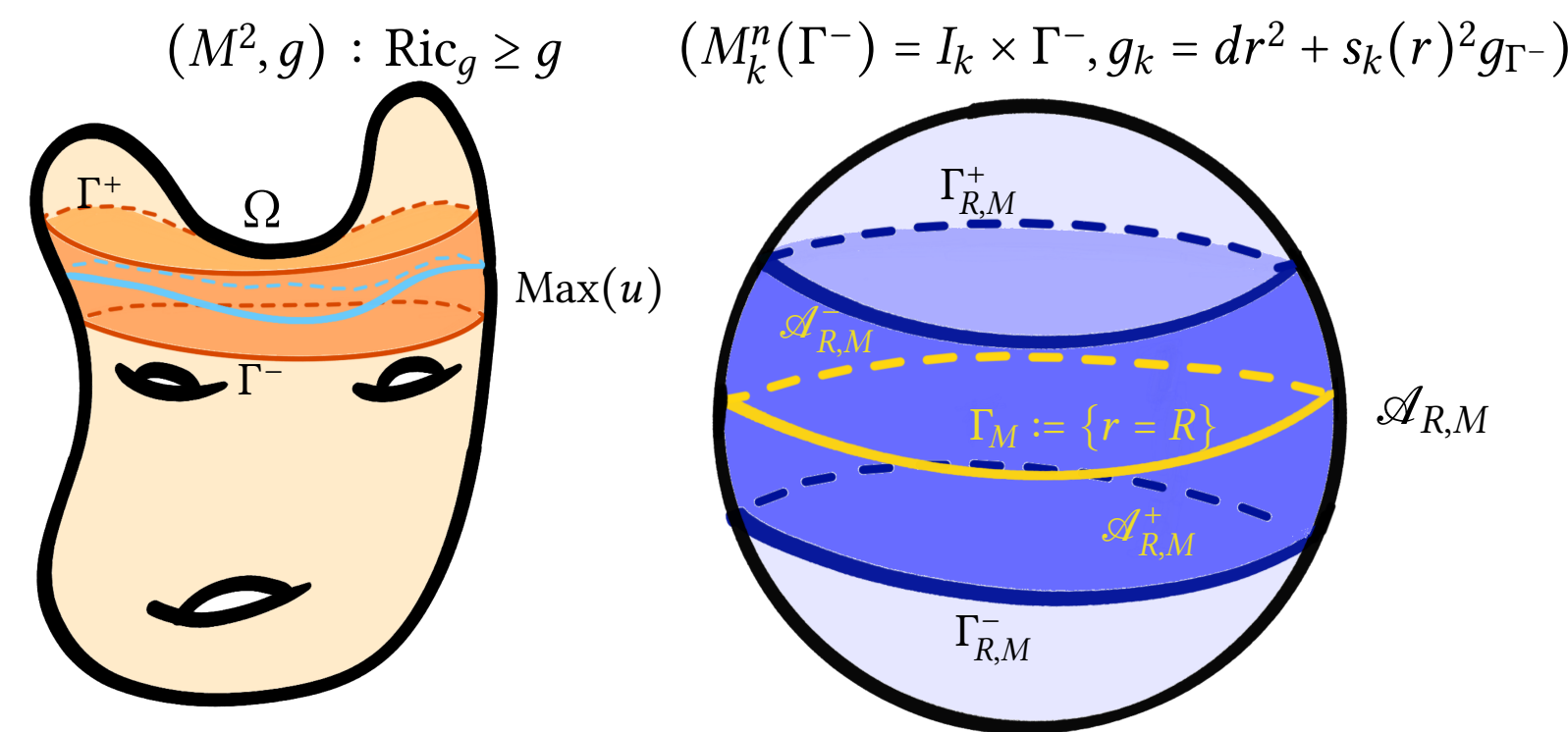
Model $\bar{r}$ -functions are defined as,

$$\bar{r}_M^\pm := \frac{V'_{R,M}(r_\pm(R, M))^2}{V'_{0,M}(r_+(0, M))^2}$$

Comparison algorithm

We fix $\mathfrak{U}$ be a connected component of $\Omega \setminus \text{Max}(u)$. For each connected component of the boundary $\Gamma \in \pi_0(\partial \mathfrak{U})$, define

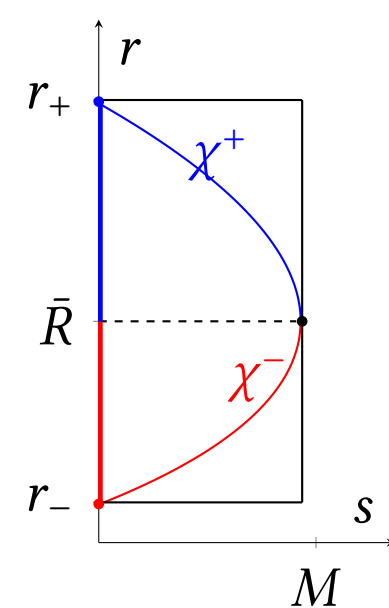
$$\begin{aligned} \bar{r}(\Gamma) &:= \max_{p \in \Gamma} \frac{|\nabla u(p)|^2}{V'_{0,M}(r_+(0, M))^2}, \quad \bar{r}(\mathfrak{U}) := \max\{\bar{r}(\Gamma) : \Gamma \in \pi_0(\partial \Omega \cap \text{cl}(\mathfrak{U}))\} \\ &\& \bar{r}(\Omega) := \max\{\bar{r}(\mathfrak{U}) : \mathfrak{U} \in \pi_0(\Omega \setminus \text{Max}(u))\}. \end{aligned}$$



We say that a model pair in $M_k^n(\Gamma_0)$ $(\bar{\mathcal{A}}, \bar{v}) := (\bar{\mathcal{A}}_{R,M}^\pm, v_{R,M}^\pm) \in \text{Model}_M^\pm$ is:

- A *comparison pair* for $(\mathfrak{U}, u)$ if $\bar{r}(\mathfrak{U}) \leq \bar{r}(\bar{\mathcal{A}}) = \bar{r}_M^\pm(R)$,
 - An *associated model pair* if $\bar{r}(\mathfrak{U}) = \bar{r}(\bar{\mathcal{A}}) = \bar{r}_M^\pm(R)$
- We fix $(\bar{\mathcal{A}}, \bar{v})$ a comparison pair with core radius $\bar{R}$ and $\bar{v}(\bar{r}_\pm) = 0$.

The comparison tool



We define the function,

$$G : [0, M] \times [\bar{r}_-, \bar{r}_+] \rightarrow \mathbb{R} \\ (s, r) \rightarrow G(s, r) := s - \bar{V}(r).$$

By the Implicit Function Theorem, there exist two functions

$$\begin{aligned} \chi^- : [0, M] &\rightarrow [\bar{r}_-, \bar{R}] \\ \chi^+ : [0, M] &\rightarrow [\bar{R}, \bar{r}_+] \end{aligned}$$

Such that $G(s, \chi^\pm(s)) = 0 \quad \forall s \in [0, M]$,
i.e. $\bar{V}(\chi^\pm(s)) = s$.

The *pseudo-radial function* associated to $(\mathfrak{U}, u)$ and $(\bar{\mathcal{A}}, \bar{v})$ is

$$\Psi : \mathfrak{U} \rightarrow \chi^\pm(0, M) \subset [\bar{r}_-, \bar{r}_+], \quad \Psi(p) := \chi^\pm(u(p)),$$

which will allow us to compare the geometry of level sets.

Notation

- $I_k := [0, \bar{r}_k)$, where $\bar{r}_k := +\infty$ if $k \leq 0$ and $\bar{r}_k := \pi/\sqrt{k}$ if $k > 0$.
- $s_k(r) := \begin{cases} \frac{\sinh(\sqrt{-k}r)}{\sqrt{-k}} & \text{if } k < 0 \\ r & \text{if } k = 0 \\ \frac{\sin(\sqrt{k}r)}{\sqrt{k}} & \text{if } k > 0 \end{cases}$ and $\cot_k(r) := \frac{s'_k(r)}{s_k(r)}$ and $\tan_k(r) = \frac{s_k(r)}{s'_k(r)}$.
- $r_\Omega = \max\{\text{dist}(p, \partial \Omega) : p \in \Omega\}$

Comparison results

Gradient estimates

Let $f \in \mathcal{C}^1(\mathbb{R})$ be a function satisfying condition:

$$f'(x) \geq nk \quad \text{for all } x \in \mathcal{I}_f. \quad (\star)$$

We have

$$|\nabla u|^2(p) \leq \bar{V}'(\Psi)^2(p) \quad \text{for all } p \in \partial \mathfrak{U}.$$

If equality holds at any point $p \in \mathfrak{U}$

- $\text{Ric}(\nabla u, \nabla u) = (n-1)k |\nabla u|^2$ on $\mathfrak{U}$,
- (M^n, g) is isometric to $(M_k^n(\Gamma_0), g_k)$ on $\mathfrak{U}$,
- $(\mathfrak{U}, u) \equiv (\bar{\mathcal{A}}, \bar{v})$.

Location of the hot-spots

Let $f \in \mathcal{C}^1(\mathbb{R})$ the condition $(\star)$ and

$$f(x) \leq f_k(x) = nkx + 1$$

Assume that Ω is contained in a convex domain of M and that $\bar{r}(\Omega) = \bar{r}(\mathfrak{U})$. If $k > 0$ suppose that $\bar{r}_+ \leq \bar{r}_k/2$. Then, for all $p \in \text{Max}(u)$:

$$\frac{\text{dist}(p, \partial \Omega)}{\tan_k(r_\Omega)/n} \begin{cases} \geq \frac{\bar{r}_+ - \bar{R}}{\sqrt{\frac{2}{n}M + kM^2}} & \text{if } (\bar{\mathcal{A}}, \bar{v}) \in \text{Model}_M^+, \\ > \frac{\bar{R} - \bar{r}_-}{\sqrt{\frac{2}{n}M + kM^2}} & \text{if } (\bar{\mathcal{A}}, \bar{v}) \in \text{Model}_M^-. \end{cases}$$

If equality holds at a point of Ω

- (M, g) is isometric to the space form $\mathbb{M}^n(k)$,
- $f(x) = nkx + 1$,
- (Ω, u) is a ball solution of Serrin's problem.

Isoperimetric type inequality

Let $f \in \mathcal{C}^m(\mathbb{R})$, with $m = \max\{n-2, 3\}$, satisfying condition $(\star)$. Assume that the $(n-1)$ -dimensional part of $\text{cl}(\mathfrak{U}) \cap \text{Max}(u)$, denoted by $\bar{\Gamma}_M$, consists of a possibly disconnected Lipschitz-continuous hypersurface. Hence,

$$\frac{\mathcal{H}^n(\mathfrak{U})}{\mathcal{H}^{n-1}(\bar{\Gamma}_M)} \geq \begin{cases} \frac{\mathcal{H}_k^n(\bar{\mathcal{A}})}{\mathcal{H}_k^{n-1}(\bar{\Gamma}_M)} & \text{if } (\bar{\mathcal{A}}, \bar{v}) \in \text{Model}_M^+, \\ \frac{\mathcal{H}_k^n(\bar{\mathcal{A}})}{\mathcal{H}_k^{n-1}(\bar{\Gamma}_M)} & \text{if } (\bar{\mathcal{A}}, \bar{v}) \in \text{Model}_M^-. \end{cases}$$

The equality holds if and only if $(\mathfrak{U}, u) \equiv (\bar{\mathcal{A}}, \bar{v})$.

Bibliography

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